THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 07

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 1296	$36 \times 36 = 1296$
2	**(a)** 23	$23 \times 23 = 529$
3	**(c)** 0.002	Move the decimal 3 places left
4	**(d)** 149	Add the values, subtracting a smaller symbol that comes before a larger one
5	**(a)** 108°	Adjacent angles are supplementary
6	**(a)** trapezium	Trapezium
7	**(b)** 1	Digit sum must be a multiple of 3
8	**(c)** 4	$xp + xq + yp + yq$
9	**(b)** ₹ 135	₹ 7.5 each
10	**(c)** ₹ 200	A gets 2/5 of 500
11	**(c)** 45%	× 100
12	**(b)** ₹ 1440	$SI = \frac{PRT}{100}$
13	**(b)** $12\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
14	**(d)** 17 cm	$\sqrt{8^2 + 15^2} = 17$
15	**(b)** inverse proportion	Definition
16	**(c)** prism	Definition
17	**(d)** 46.4286	Sum 325 ÷ 7
18	**(b)** 121	$11(a + b)$
19	**(c)** 40 cm²	$\frac{1}{2}d_1d_2$
20	**(d)** 198 cm²	$l \times b$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$\frac{1}{2^6} = 2^{-6}$ (2).	2
22	Al-Khwarizmi and Al-Kindi used the name Hindu numerals (1); Fibonacci promoted them in Europe (1).	1+1
23	Every other even number has 2 as a factor besides 1 and itself (2).	2
24	$45 : 1440 = 1 : 32$ (2).	2
25	$5\sqrt{2}$ cm (2).	2
26	15 cm (2).	2
27	Total 54 then 70; number 16 (2).	2
28	9 cm (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	1, 8, 27, 64, 125 (2); odd cubes are 1, 27, 125 (1).	2+1
30	Perimeter $= 2(12 + 9) = 42$ cm (1.5); area needs the height, not just sides (1.5).	1½+1½
31	$x^2 + 14x + 48$ (2); at $x = 1$: $(7)(9) = 63 = 1 + 14 + 48$ (1).	2+1
32	SI $=$ ₹480 (2); amount $=$ ₹2880 (1).	2+1
33	$2(40 + 24 + 15) = 158$ cm² (3).	3
34	Choose equal or closest numbers (1); 15 and 15 (1); product 225 (1).	1+1+1
35	By the theorem, height $= \sqrt{20^2 - 12^2}$ (2); $= 16$ m (1).	2+1
	OR: Diagonal 13 m (1.5); saved $17 - 13 = 4$ m (1.5).	1½+1½

36	Trapezium $= \frac{1}{2}(18+30) \times 12 = 288$ cm² (2); rectangle 360 cm² (1).	2+1
	OR: Area 324 m² (1.5); perimeter 72 m, cost ₹ 2160 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $2^5 = 32$ (2). (ii) $(-2)^{3-5+1} = (-2)^{-1} = -\frac{1}{2}$ (3).	2+3
	OR: (i) $6^3 = 216 = 8 \times 27$ (2). (ii) $(12/6)^3 = 8$ (3).	2+3
38	(i) Definitions (2). (ii) SP ₹ 2200; GST ₹ 396; total ₹ 2596 (3).	2+3
	OR: (i) ₹ 25500 (2). (ii) ₹ 24000 (1). (iii) $6000/30000 = 20\%$ (2).	2+1+2
39	(i) $4 \times 12 = 48$; $48 - 32 = 16$ (3). (ii) Sorted 8, 10, 14, 16; median $(10+14)/2 = 12$ (2).	3+2
	OR: (i) 42 (2). (ii) For 17 numbers, the median is the 9th value (3).	2+3
40	(i) Both equal base × common height (3). (ii) Base 10, height 6 gives 60 for both (2).	3+2
	OR: (i) 10×10=100, 12×8=96, 15×5=75, 18×2=36 (3). (ii) 10 × 10 square (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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